Project Title
E-Commerce Product Entity-Resolution & Matching Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The E-Commerce Product Entity-Resolution & Matching Platform is designed to identify and match listings that refer to the same physical product across multiple online sellers. Product titles and attributes often vary because of differences in naming, formatting, abbreviations, and seller descriptions, making direct comparison difficult. The proposed system preprocesses product information, extracts important attributes, and applies efficient string-matching, hashing, and similarity-based techniques to detect equivalent listings. Matched listings are grouped under a canonical product representation, enabling users to compare the same product across different sellers and prices. The project demonstrates the practical application of data structures and algorithms for fast searching, matching, deduplication, and organization of e-commerce data. It aims to improve product discovery, reduce duplicate catalogue entries, and support accurate price comparison.

Signature of the Guide